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ME 362 VIBRATIONS I

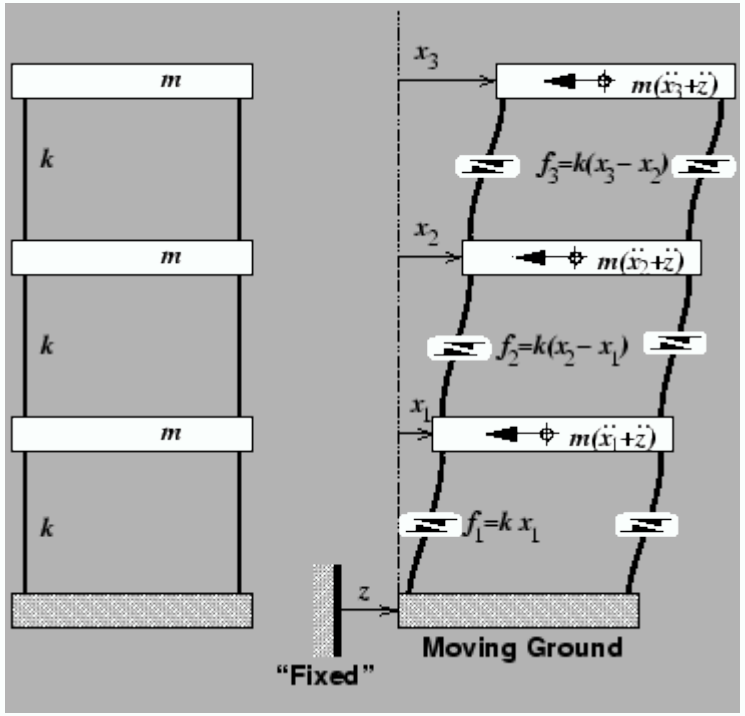
Lecture 3

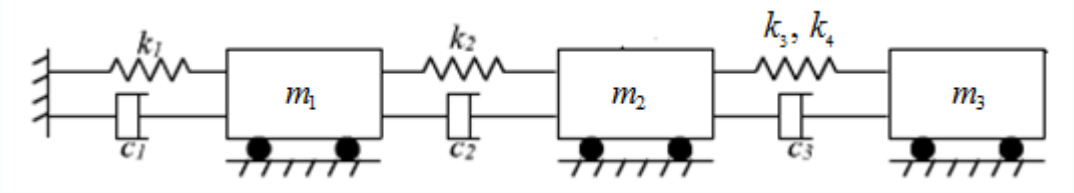
VIBRATION OF TWO-DEGREE-OF-FREEDOM SYSTEMS

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Models


m
k
m
k
m
k
x3
m(x3+z)
f3=k(x3− x2)
x2
m(x2+z)
f2=k(x2− x1)
x1
m(x1+z)
f1=k x1
z
Moving Ground
"Fixed"


k1
k2
k3, k4
m1
m2
m3
c1
c2
c3



Systems with more than one degree of freedom

- Many real systems can be represented by a single degree of freedom model.
- However, most actual systems have several bodies and several restraints and therefore several degrees of freedom.
- Since no body is completely rigid, and no spring is without mass, every real system has more than one degree of freedom, and sometimes it is not sufficiently realistic to approximate a system by a single degree of freedom model.
- Thus, it is necessary to study the vibration of systems with more than one degree of freedom.

General Solution-Free Undamped Systems

$$[m]\{\ddot{x}\}+[k]\{x\}=\{0\}$$

$$x = A\sin\omega t$$

$$let\ \lambda = \omega^2$$

$$[k/m-\lambda]\{x\}=\{0\}$$

$$let\ D = k/m$$

$$|[D]-\lambda[I]|=0$$

$$\begin{vmatrix} D_1-\lambda & 0 & . & . & 0 \\ 0 & D_2-\lambda & 0 & . & . \\ 0 & . & . & 0 & 0 \\ 0 & . & . & . & 0 \\ 0 & . & . & . & D_n-\lambda \end{vmatrix} = 0$$

Mode Shapes – Principal Coordinates

$$[[D]-\lambda_i[I]]\{x_i\}=\{0\}$$

$$p=\left[\begin{Bmatrix} x_1 \\ x_2 \\ . \\ x_n \end{Bmatrix}_1 \begin{Bmatrix} x_1 \\ x_2 \\ . \\ x_n \end{Bmatrix}_2 \ldots \begin{Bmatrix} x_1 \\ x_2 \\ . \\ x_n \end{Bmatrix}_n\right]$$

Normalised Weighed Modal Matrix

$$\bar{p}=\frac{1}{\sqrt{\{x\}^T[m]\{x\}}}p$$

General Solution-Free Undamped Systems
Natural Frequency and mode shapes

$$[m]\{\ddot{x}\} + [k]\{x\} = \{0\}$$

$$\ddot{x} = -\omega^2 x = -\lambda x$$

$$\left[[k][m]^{-1} - \lambda\right]\{x\} = \{0\}$$

$$let\ \ D = [k][m]^{-1};\ \text{Dynamic Matrix}$$

$$|[D] - \lambda[I]| = 0$$

Mode Shapes- Principal Coordinates

$$([D] - \lambda_i[I])\{x_i\} = \{0\}$$

Modal Matrix p,

$$p = \left[\begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix}_1 \quad \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix}_2 \quad \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix}_3 \ \ldots\right]$$

General Solution-Forced Damped Systems

$$[m]\{\ddot{x}\}+[c]\{\dot{x}\}+[k]\{x\}=\{F\}$$

$$\{x\}=[\bar{p}]\{y\}$$

$$[\bar{p}]^T[m][\bar{p}]\{\ddot{y}\}+[\bar{p}]^T[c][\bar{p}]\{\dot{y}\}+[\bar{p}]^T[k][\bar{p}]\{y\}=[\bar{p}]^T\{F\}$$

$[c]=\alpha[M]+\beta[K]$; Rayleigh Proportional Damping

$$\begin{bmatrix}1&0&.&.&0\\0&1&0&.&.\\0&.&.&0&0\\0&.&.&.&0\\0&.&.&.&1\end{bmatrix}\{\ddot{y}\}+\begin{bmatrix}a_1&0&.&.&0\\0&a_2&0&.&.\\0&.&.&0&0\\0&.&.&.&0\\0&.&.&.&a_n\end{bmatrix}\{\dot{y}\}+\begin{bmatrix}b_1&0&.&.&0\\0&b_2&0&.&.\\0&.&.&0&0\\0&.&.&.&0\\0&.&.&.&b_n\end{bmatrix}\{y\}=\begin{Bmatrix}c_1\\c_2\\.\\.\\c_n\end{Bmatrix}$$

General Solution

$$\ddot{y}_1 + a_1 \dot{y}_1 + b_1 y_1 = c_1$$

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$$\ddot{y}_n + a_n \dot{y}_n + b_n y_n = c_n$$

$$\{x\} = \left[\overline{p}\right]^{-1} \{y\}$$

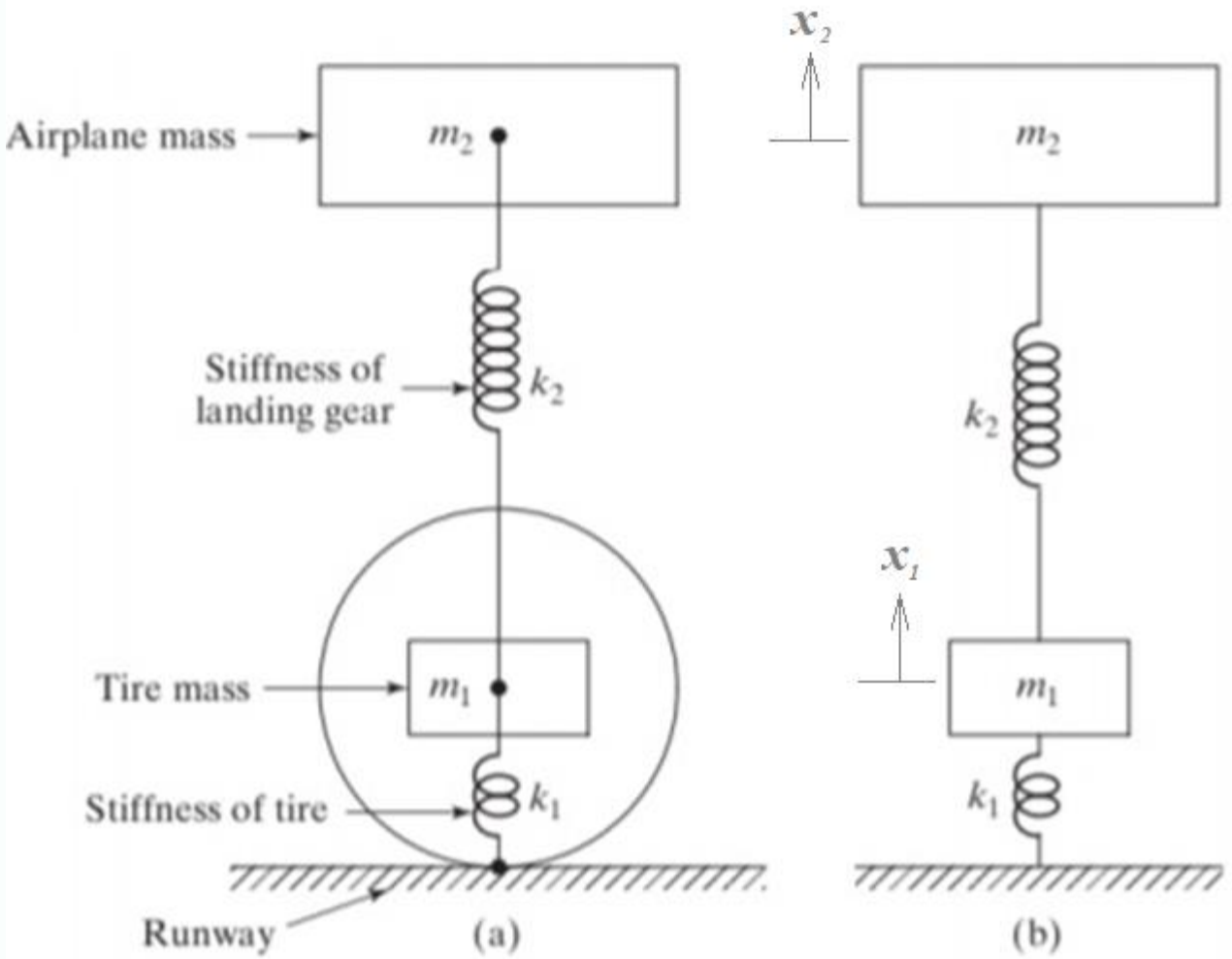
Example 1

A simplified model of the main landing gear system of a small airplane is shown with

$$m_1 = 9\ kg;\ m_2 = 1\ kg;$$
$$k_1 = 24\ N/m; and\ \ k_2 = 3\ N/m$$

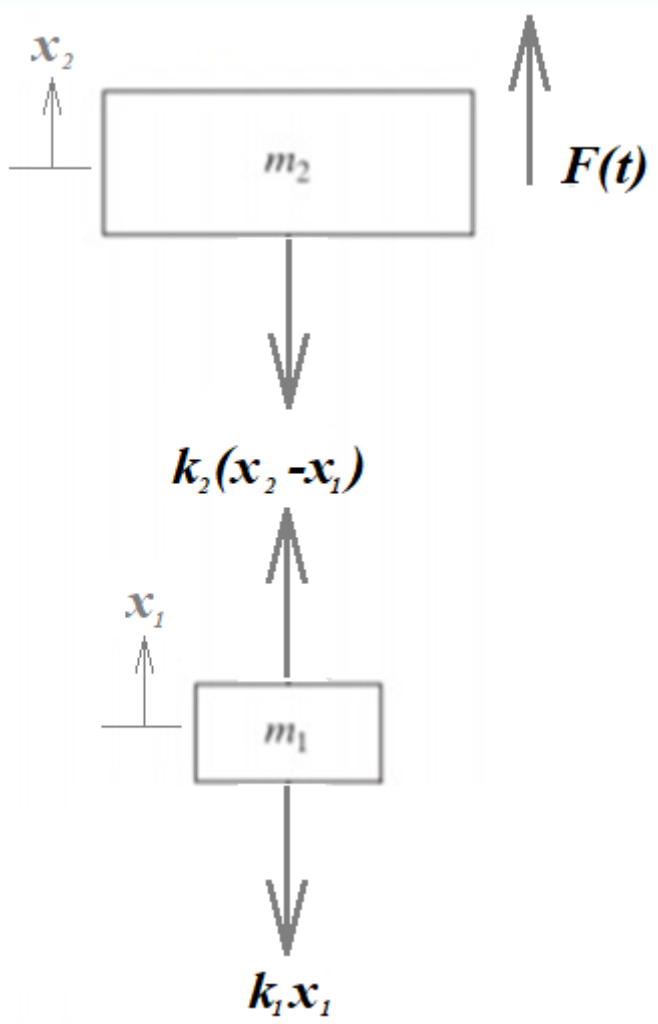
1. Write the equation of motion of the system.
2. Find the natural frequency and the mode shapes of the system.
3. Find the free response of the system.
4. Find the response if a vertical harmonic force of 3 cos(3t) N acts on the Airplane mass.

$$x(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{mm},\ \dot{x}(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



Example 1

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{Bmatrix} + \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} =$$



Let $\lambda=\omega^2$

$$\begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix} \ddot{\mathbf{x}} + \begin{bmatrix} 27 & -3 \\ -3 & 3 \end{bmatrix} \mathbf{x} = \mathbf{0}$$

$$\begin{vmatrix} 27 - 9\lambda & -3 \\ -3 & 3 - \lambda \end{vmatrix} = 0;$$

λ=2, 4

Natural frequencies

$$\omega_1 = \sqrt{2} = 1.414\, rad/s$$

$$\omega_2 = \sqrt{4} = 2\, rad/s$$

0

Example 1

First mode, λ=2

$$\begin{bmatrix} 27-9(2) & -3 \\ -3 & 3-(2) \end{bmatrix} \begin{Bmatrix} C_1 \\ C_2 \end{Bmatrix} = 0;$$

$$\frac{C_2}{C_1} = \frac{3}{1}$$

Second mode

$$\begin{bmatrix} 27-9(4) & -3 \\ -3 & 3-(4) \end{bmatrix} \begin{Bmatrix} C_1 \\ C_2 \end{Bmatrix} = 0;$$

$$\frac{C_2}{C_1} = -\frac{3}{1}$$

Modal matrix, p

$$p = \begin{bmatrix} 1 & 1 \\ 3 & -3 \end{bmatrix}$$

Example 1

$$\begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix} \ddot{\mathbf{x}} + \begin{bmatrix} 27 & -3 \\ -3 & 3 \end{bmatrix} \mathbf{x} = \mathbf{0}$$

$$[m]\{\ddot{x}\} + [k]\{x\} = \{0\}$$

Let $\{x\} = [p]\{y\}$

$$[p]^T[m][p]\{\ddot{y}\} + [p]^T[k][p]\{y\} = \{0\}$$

$$[p]^T[m][p] = \begin{bmatrix} 18 & 0 \\ 0 & 18 \end{bmatrix}$$

$$[p]^T[k][p] = \begin{bmatrix} 36 & 0 \\ 0 & 72 \end{bmatrix}$$

$$\ddot{y}_1 + 2y_1 = 0 \quad and \quad \ddot{y}_2 + 4y_2 = 0$$

$$\{y\} = \begin{Bmatrix} Asin(\sqrt{2}\,t + \phi_1) \\ Bsin(2t + \phi_2) \end{Bmatrix}$$

Example 1

$$\{x\} = [p]\{y\}$$

$$\{x\} = \begin{bmatrix} 1 & 1 \\ 3 & -3 \end{bmatrix} \begin{Bmatrix} A\sin(\sqrt{2}\,t + \phi_1) \\ B\sin(2t + \phi_2) \end{Bmatrix}$$

$$\{x\} = \begin{Bmatrix} A\sin(\sqrt{2}\,t + \phi_1) + B\sin(2t + \phi_2) \\ 3A\sin(\sqrt{2}\,t + \phi_1) - 3B\sin(2t + \phi_2) \end{Bmatrix}$$

$$\{\dot{x}\} = \begin{Bmatrix} \sqrt{2}\,A\cos(\sqrt{2}\,t + \phi_1) + 2B\cos(2t + \phi_2) \\ 3\sqrt{2}\,A\cos(\sqrt{2}\,t + \phi_1) - 6B\cos(2t + \phi_2) \end{Bmatrix}$$

Applying initial conditions gives

$$A = B = 0.5\,mm, \phi_1 = \phi_2 = \pi/2$$

$$\therefore \{x\} = \begin{Bmatrix} 0.5\sin(\sqrt{2}\,t + \pi/2) + 0.5\sin(2t + \pi/2) \\ 1.5\sin(\sqrt{2}\,t + \pi/2) - 1.5\sin(2t + \pi/2) \end{Bmatrix} mm$$

Example 1: Forced response

$$[m]\{\ddot{x}\} + [k]\{x\} = \{F\}$$

Let $\{x\} = [p]\{y\}$

$$[p]^T[m][p]\{\ddot{y}\} + [p]^T[k][p]\{y\} = [p]^T\{F\}$$

$$\ddot{y}_1 + 2y_1 = 9\cos 3t \quad and \quad \ddot{y}_2 + 4y_2 = -9\cos 3t$$

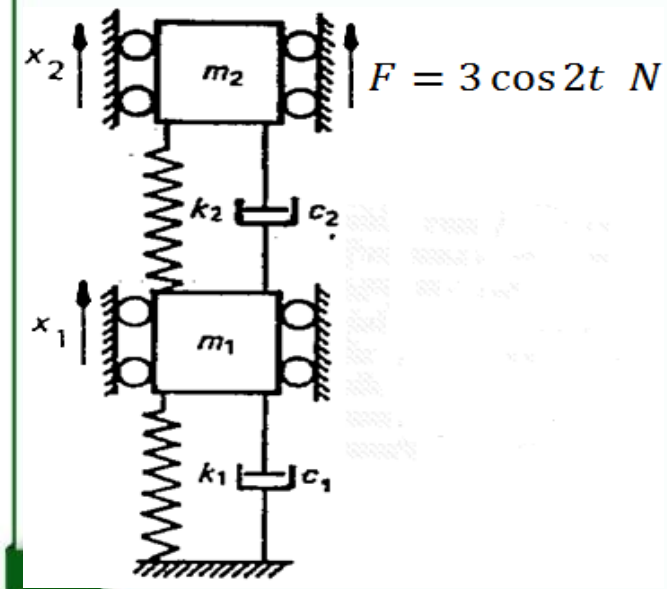
$$\{y\} = \begin{Bmatrix} Asin\left(\sqrt{2}\,t + \phi_1\right) - 1.286\cos 3t \\ Bsin(2t + \phi_2) + 1.8\cos 3t \end{Bmatrix}$$

Example 2

Consider the simple two-degree-of-freedom system with a harmonic force applied to one mass as indicated in the Figure. For this example, let m_1 = 9 kg, m_2 = 1 kg, k_1 = 24 N/m, k_2 = 3 N/m, c_1 = 2.4 N· s/m and c_2 = 0.3 N·s/m. Calculate the Steady- State Response

F.B.D ? Equations of motion ?

$$\begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix} \ddot{\mathbf{x}} + \begin{bmatrix} 2.7 & -0.3 \\ -0.3 & 0.3 \end{bmatrix} \dot{\mathbf{x}} + \begin{bmatrix} 27 & -3 \\ -3 & 3 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 0 \\ 3cos2t \end{bmatrix}$$



i.

Steady- State Response

Modal Matrix, p

$$p = \begin{bmatrix} 1 & -1 \\ 3 & 3 \end{bmatrix}$$

$$\{x_1\}'[m]\{x_1\} = \begin{bmatrix} 1 & 3 \end{bmatrix} \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = 18$$

$$\{x_1\}'[k]\{x_1\} = \begin{bmatrix} 1 & 3 \end{bmatrix} \begin{bmatrix} 27 & 3 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = 18$$

Normalised Weighed Modal Matrix

$$\bar{p} = \frac{1}{\sqrt{\{x\}^T [m]\{x\}}} p$$

$$\bar{p} = \frac{1}{\sqrt{18}} \begin{bmatrix} 1 & -1 \\ 3 & 3 \end{bmatrix}$$

Solution

$$[m]\{\ddot{x}\}+[c]\{\dot{x}\}+[k]\{x\}=\{F\}$$

$$\{x\}=[\bar{p}]\{y\}$$

$$[\bar{p}]^T[m][\bar{p}]\{\ddot{y}\}+[\bar{p}]^T[c][\bar{p}]\{\dot{y}\}+[\bar{p}]^T[k][\bar{p}]\{y\}=[\bar{p}]^T\{F\}$$

$$[\bar{p}]^T[m][\bar{p}]=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

$$[\bar{p}]^T[k][\bar{p}]=\begin{bmatrix}2 & 0\\ 0 & 4\end{bmatrix}$$

$$[\bar{p}]^T[c][\bar{p}]=\begin{bmatrix}0.2 & 0\\ 0 & 0.4\end{bmatrix}$$

$$[\bar{p}]^T\{F\}=2.12\cos 2t\begin{Bmatrix}1\\ 1\end{Bmatrix}$$

$$\ddot{y}_1+0.2\dot{y}_1+2y_1=2.12\cos 2t$$

$$\ddot{y}_1+0.4\dot{y}_1+4y_1=2.12\cos 2t$$

$$y_1=\frac{\delta}{\sqrt{\left(1-\tau^2\right)^2+\left(2\xi\tau\right)^2}}$$

$$y_1=1.04\cos(2t+0.198)$$

$$y_2=2.65\cos(2t+\pi/2)$$

Solution

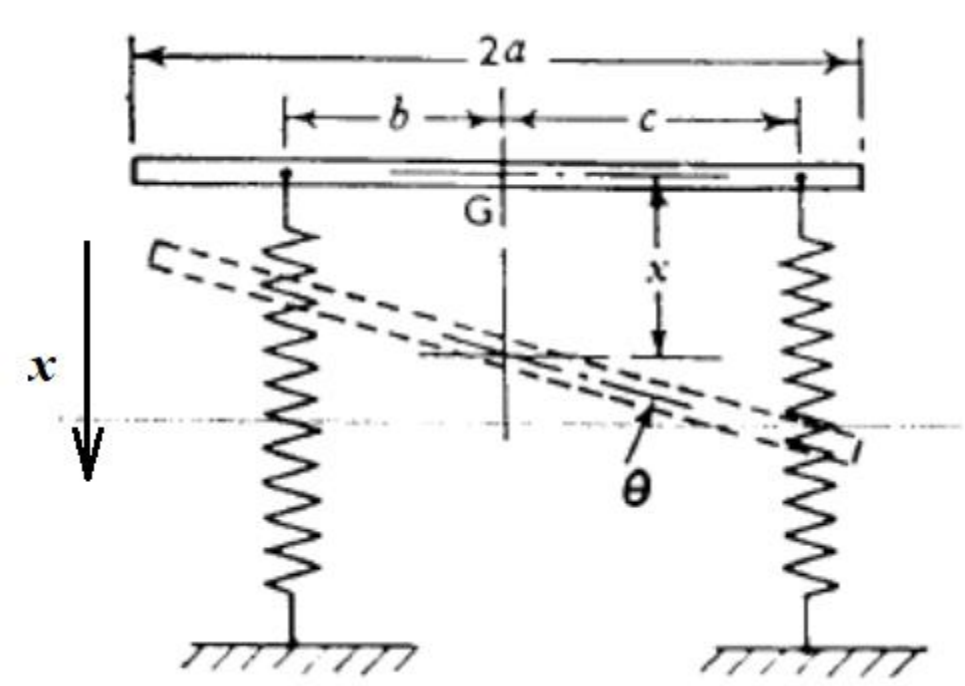
$$\{x\} = [\overline{p}]\{y\}$$

$$\{x\} = \frac{1}{\sqrt{18}}\begin{bmatrix} 1 & -1 \\ 3 & 3 \end{bmatrix}\begin{Bmatrix} 1.04\cos(2t + 0.198) \\ 2.65\cos(2t - \pi/2) \end{Bmatrix}$$

$$x_1 = 0.245\cos(2t + 0.198) - 0.625\sin 2t$$

$$x_2 = 0.736\cos(2t + 0.198) + 1.875\sin 2t$$

Vibration of two-degree-of-freedom systems



A uniform beam of length 2a and mass m *is supported horizontally on two equal springs, each of stiffness S, at points distant band* c *from the middle. Derive the equation of the two frequencies of small oscillations in the vertical plane. Show that these frequencies are equal only if* $b=c=a/\sqrt{3}$.

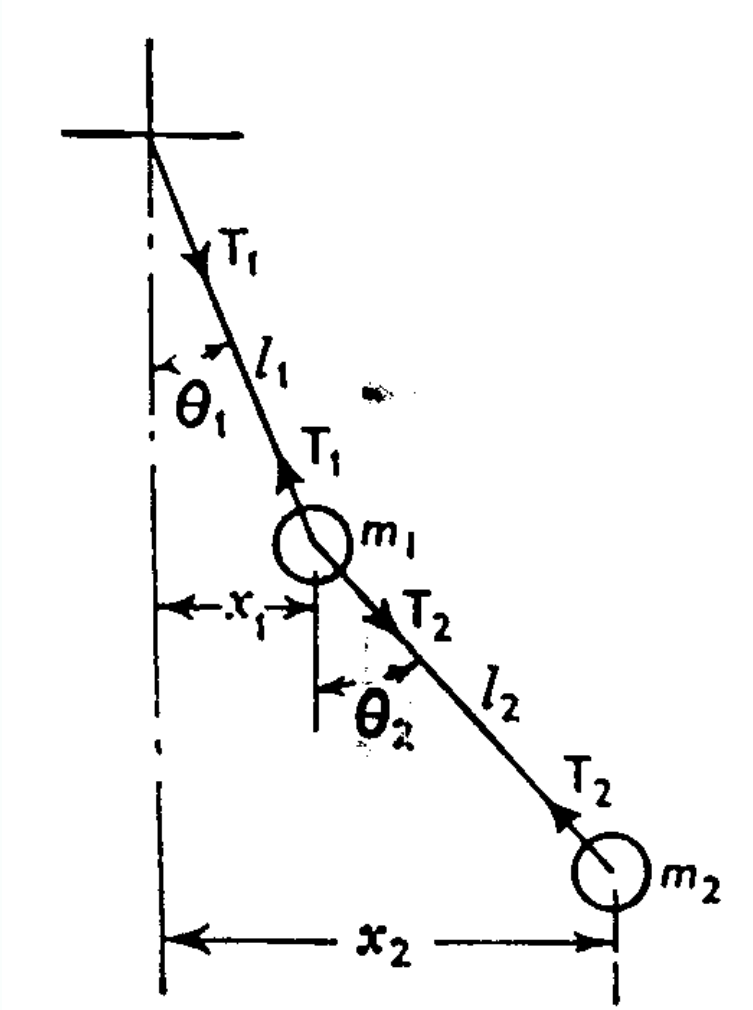
$$m\ddot{x} = -k(x - b\theta) - k(x + c\theta)$$

The moment about G

$$\mathrm{I}\ddot{\theta} = k(x - b\theta)b - k(x + c\theta)c$$

$$I = \frac{m(2a)^2}{12}$$

Vibration of two-degree-of-freedom systems



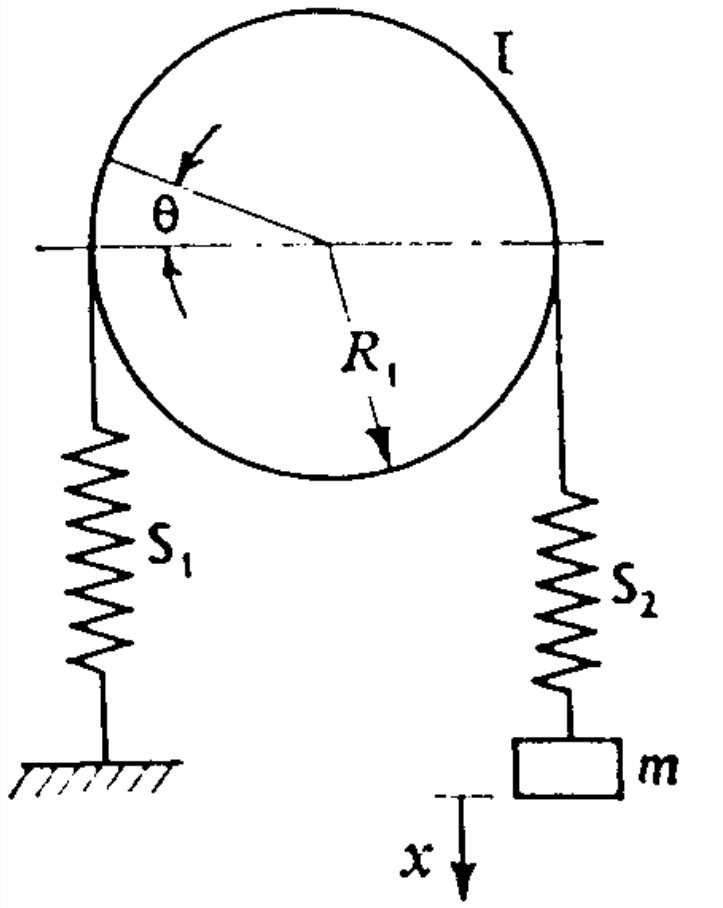
$$m_1\ddot{x}_1 = -T_1\theta_1 + T_2\theta_2$$

$$m_2\ddot{x}_2 = -T_2\theta_2$$

$$x_1 \cong l_1\theta_1 \quad and \quad x_2 - x_1 \cong l_2\theta_2$$

$$T_2 \cong m_2 g \quad and \quad T_1 \cong m_1 g + T_2$$

Vibration of two-degree-of-freedom systems

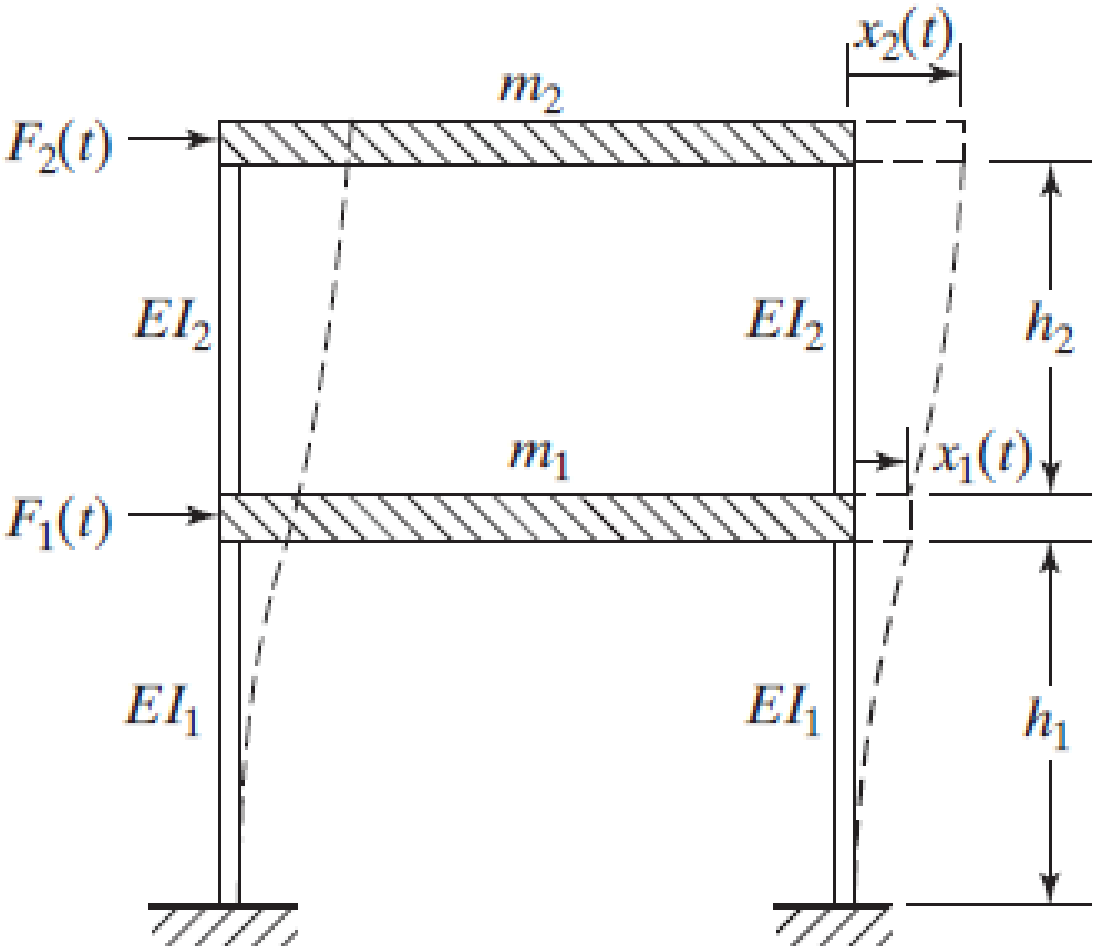


$$m\ddot{x} = -S_2(x - R_1\theta)$$

The moment about G

$$I\ddot{\theta} = S_2(x - R_1\theta)R_1 - S_1{R_1}^2\theta$$

Vibration of two-degree-of-freedom systems



$$m_1\ddot{x}_1 = F_1 + k_2(x_2 - x_1) - k_1x_1$$

$$m_2\ddot{x}_2 = F_2 - k_2(x_2 - x_1)$$

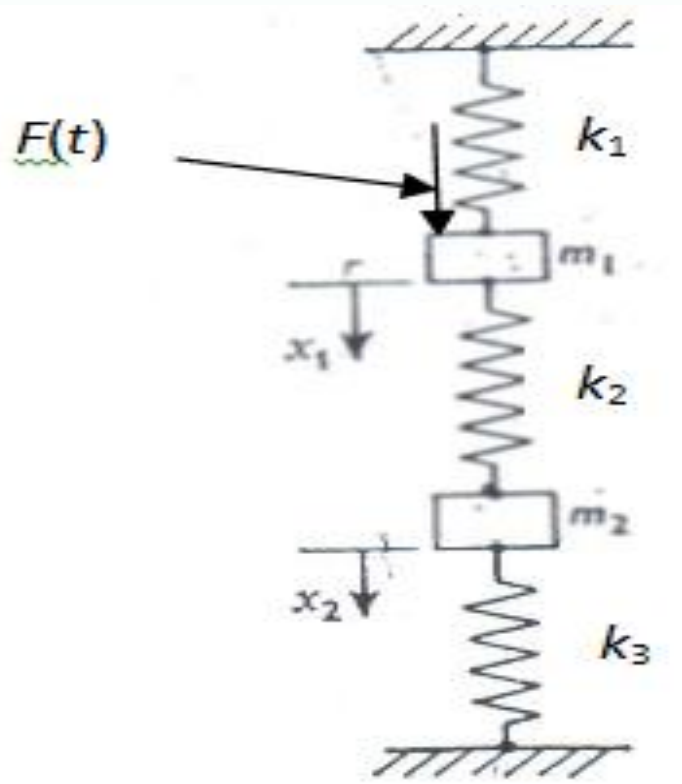
Assignment 1

The Figure shown, shows a system consisting of two bodies and three springs with a harmonic force

$F(t) = 15 \cos 3t$ kN, applied to one mass. Determine the natural frequencies of this system and calculate its steady-state response. Obtain the solution in the physical coordinate system $\mathbf{x}(t)$. Work from first principles using matrix methods.

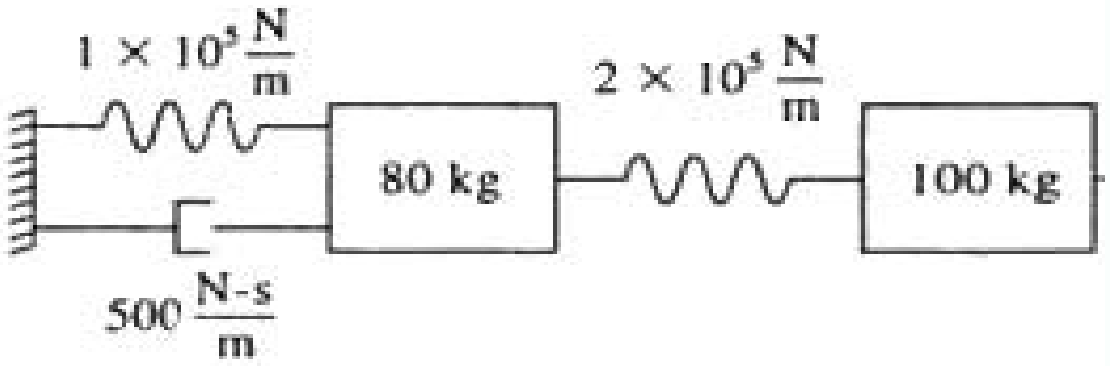
Take m_1 =7 kg, m_2 =17kg, k_1 = 30 kN/m, k_2 = 20 kN/m, and k_3 = 16 kN/m, where k denotes spring constant.

$$x(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} cm,\ \dot{x}(0) = \begin{bmatrix} -2 \\ 0 \end{bmatrix} \text{cm/s}$$



Assignment 2

Find the natural frequency and the Free response of the system shown below

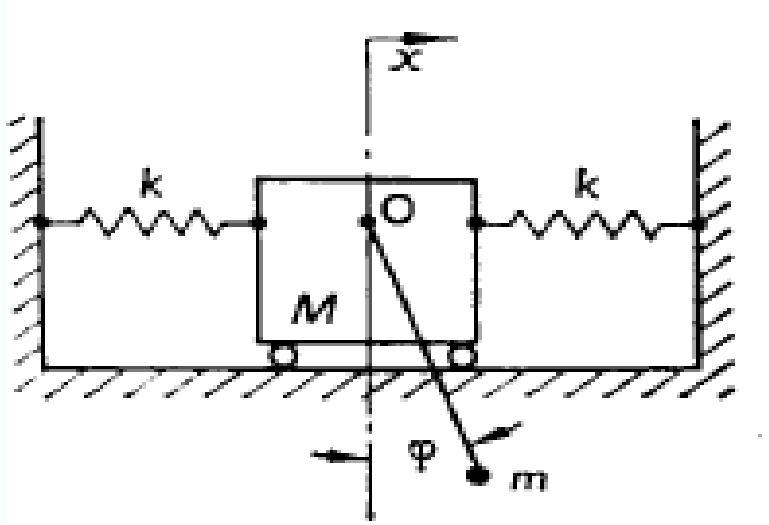


$$x(0) = \begin{bmatrix} 1 \\ 3 \end{bmatrix} cm,\ \dot{x}(0) = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \text{cm/s}$$

Assignment 3

The figure shows a trolley of mass M, which runs on a frictionless horizontal plane. At O the trolley carries a simple pendulum of length l with a body of mass m at its end.

Two equal springs, each of stiffness k, are attached to the trolley and to the fixed walls. By using the independent co-ordinates x and φ as shown in the figure, determine, for the small free oscillations, the equations of motion.



$$(M+m)\ddot{x} + 2kx + ml\ddot{\varphi} = 0$$

$$\ddot{x} + l\ddot{\varphi} + g\varphi = 0$$

If M=100 kg, m=10 kg, k= 2 kN/m, l= 2 m. Determine the natural frequencies and the response of the system shown.

$$x(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} cm,\ \dot{x}(0) = \begin{bmatrix} -2 \\ 0 \end{bmatrix} \text{cm/s}$$